04-02-24
1st class

Chemistry of natural products - 332F (Suk)

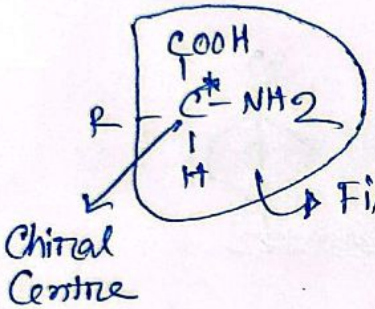
Tenpenoids | Carbohydrate → Alkaloids | Lipid → Antibiotics | Protein | Nucleic acid + Purin

book → Denis R Fennies/Fihan vol(II)
Morrison and Boyd

## ⊛ Amino acid and proteins:-

↳ Polymers of Amino acid

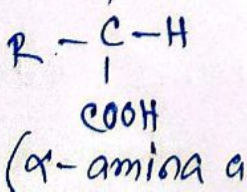
A.A



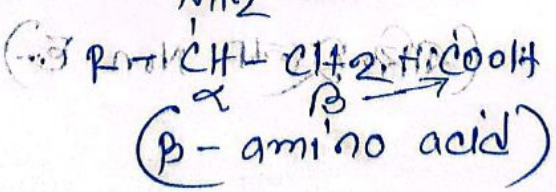

R = Carbon

Fixed (Optically active) → Plane polarized light rotate করতে পারে,
without Glycine
Chiral molecule
(Pro-chiral → 0. Inactive)

⇒ Amino acid is a molecule consist of a basic amino group (-NH3) and acydic carboxyl group (-COOH) and an organic (-R) group. Each molecule contains a central carbon atom called α-carbon, to which both an amino and a carbonyl group attached. The amino acids different from each other in the particular chemical structure of R.




(α-amino acid)




(β-amino acid)

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Except
↳ Protein and hydroxy protein contains iminoacid

(NH) all are alpha amino carboxylic acid

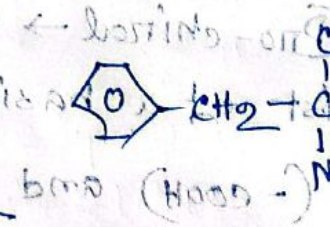


Prolin and hydroxy prolin → imino acid (2021 with structure)

R=H, $H-\underset{NH_2}{\overset{}{C}}(H)-COOH$ → Glycine



R=$CH_3$, $CH_3-\underset{NH_2}{\overset{}{C}}(H)-COOH$ → Alanine



→ Phenyl alanine

R=$(CH_3)_2-CH_2$, $(CH_3)_2-CH-\underset{NH_2}{\overset{}{C}}(H)-COOH$ → Valine

⇒ Essential Amino acid (10টি) → (Valine, Lysine, Tryptophan, Histidine)

Non- " " " (বাকি গুলো Non-E...)

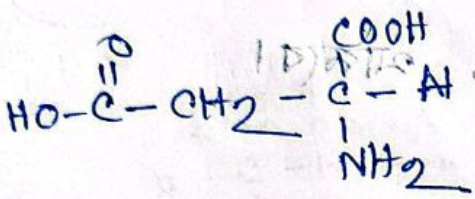
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⇒ Fundamentals of molecular spectroscopy by CN. Banwell

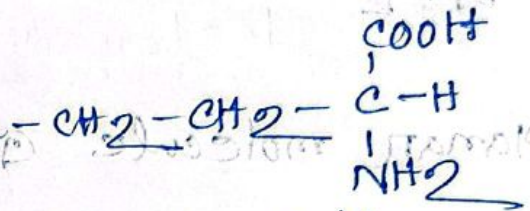
332-Sukh
2nd class

Classification of Amino Acid according to their properties of their side chain (2024)

(i) Acidic side chain



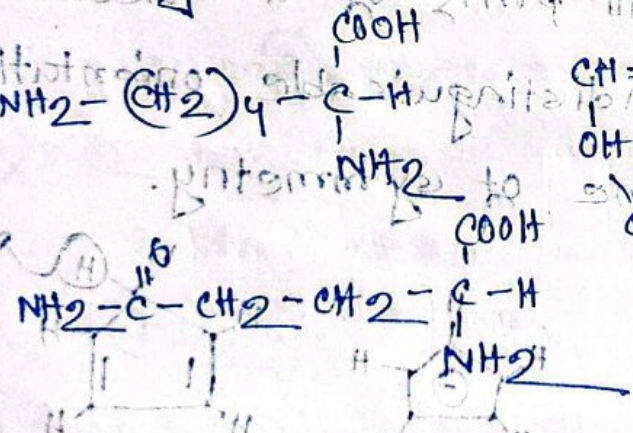
Aspartic acid



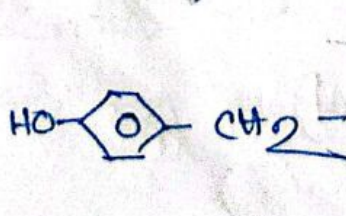
Glutamic acid

(ii) Basic side chain

Basic amino acid have side chain with an amino group which can accept proton.



(iii) Amino acid with unchanged polar group

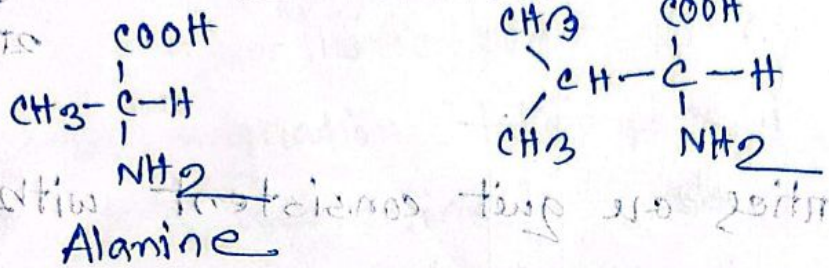


Tynosine → Blood pressure up-down কম control করে।

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(iv) Non-polar side chain

A.A has nonpolar side chain and does not gain or loss protons :



Alanine

⇒ α-amino acid গুলো organic molecule হলেও properties গুলো inorganic molecule এর কাছে মিলে

(vi) Properties of Amino Acid

(i) They are colorless crystaline solid.

(ii) They can be soluble in water but insoluble in organic solvent.

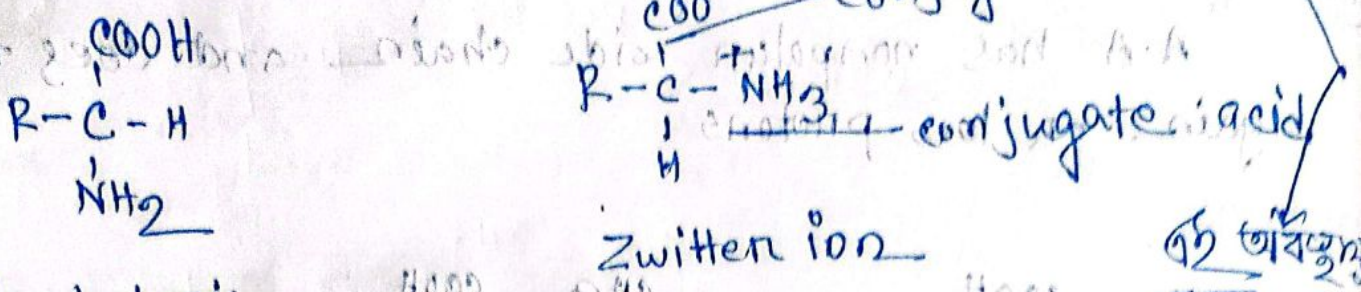
(iii) Some are volatile.

(iv) Aqueous solution behave like solutions of substance with high dipole moment.

(v) Melt with decomposition at fairly high temperature. (M.P. high)

(vi) Basicity and acidity constants are ridiculously low for $-COOH$ and $-NH_2$ group.

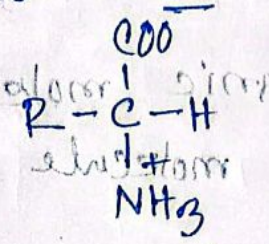
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Amphoteric compound

Zwitter ion

All this properties are quit consistent with a dipolar structure.




Dipolar ion / Zwitter ion

⇒ Amino acid is amphoteric in nature. (CT)

$$^{+}NH_3 - CHR - COO^{-} + H_2O \rightleftharpoons H_3O^{+} + NH_2 - CHR - COO^{-}$$

$$K_a = \frac{[H_3O^{+}]\,[NH_2 - CHR - COO^{-}]}{[^{+}NH_3 - CHR - COO^{-}]}$$

$$^{+}NH_3 - CHR - COO^{-} + H_2O \rightleftharpoons NH_3 - CHR - COOH + OH^{-}$$

$$K_b = \frac{[OH^{-}]\,[^{+}NH_3 - CHR - COOH]}{[^{+}NH_3 - CHR - COO^{-}]}$$

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So, A·A can accept and donate proton thus acts as an amphoteric in nature.

④ Isoelectronic point of amino acid :- 2022

(pI)

The isoelectronic point of A·A is the pH at which no net migration takes place in an electric field. It is also defined as the point at which A·A has no net electrical charge.

pI of glycine is 5.97

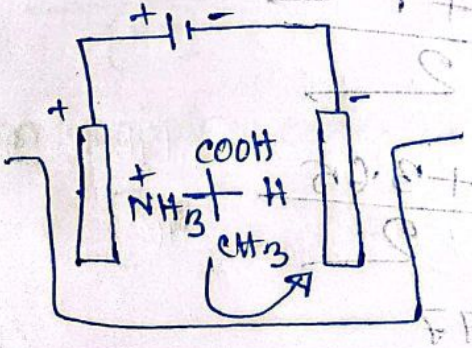
pI of leucine is 9.74

In acidic solution A·A stays as a cation and in basic medium as anion.

$$R-\underset{\overset{+}{N}H_3}{\underset{|}{CH}}-COOH \xrightleftharpoons{-H^+} R-\underset{\overset{+}{N}H_3}{\underset{|}{CH}}-COO^- \xrightleftharpoons{OH^-} R-\underset{NH_2}{\underset{|}{CH}}-COO^-$$

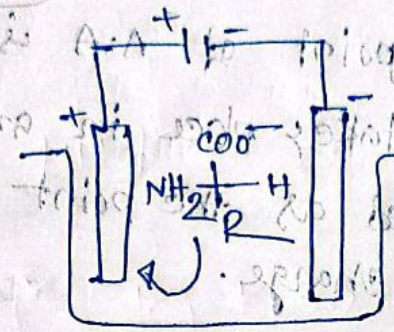
$pH < pI$ $\qquad$ $pH = pI$ $\qquad$ $pH > pI$

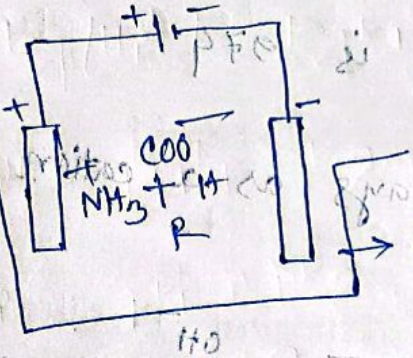
⑤ $\overset{+}{N}H_3-\underset{CH_3}{\underset{|}{C}}(COO^-)-H \xrightarrow{pH \to 0} \overset{+}{N}H_3-\underset{CH_3}{\underset{|}{C}}(COOH)-H$




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(b) $^{+}NH_3-\underset{R}{\overset{H}{C}}-COO^{-}$ $\xrightarrow{pH \to 14}$ $NH_2-\underset{R}{\overset{H}{C}}-COO^{-}$





(c)



No migration

(*) $HO-\overset{H}{\underset{H}{C}}-CH_2-\underset{+NH_3}{CH}-\overset{O}{\overset{\|}{C}}-OH$

$pK_a = 3.65$ $\quad pK_a = 1.88$

$$pI = \frac{pK_1 + pK_2}{2}$$

$$= \frac{1.88 + 3.65}{2}$$

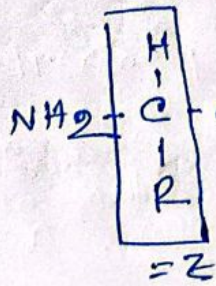
$$= 2.77$$

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19-02-24 — 3rd class

(C1) 2022

(*) Calculation of isoelectric point :-




$\equiv NH_2 - Z - COOH \equiv NH_3^+ - Z - COO^-$

$NH_3^+ - Z - COOH \underset{K_1}{\overset{H^+}{\rightleftharpoons}} NH_3^+ - Z - COO^- \underset{K_2}{\overset{OH^-}{\rightleftharpoons}} NH_2 - Z - COO^-$

Conjugate acid — C.A — $pH \sim 0.7$

Dipolan ion — DI — $pH = ?$

Conjugate base — C.B — $pH \sim 7.14$

$$K_1 = \frac{[DI][H^+]}{[CA]}$$

$$[CA] = \frac{[DI][H^+]}{K_1} \quad \text{(I)}$$

Dissociation constant

$$K_2 = \frac{[CB][H^+]}{[DI]}$$

$$[CB] = \frac{K_2[DI]}{[H^+]}$$

In isoelectric point concentration of

$[CA] = [CB]$ → (Condition)

→ তাহলে More করবেনা।

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$$[CA] = [CB]$$

$$\Rightarrow \frac{[DI][H^+]}{K_1} = \frac{K_2[DI]}{[H^+]}$$

$$\Rightarrow K_1 K_2 = [H^+]^2$$

$$\Rightarrow \log [H^+]^2 = \log K_1 K_2$$

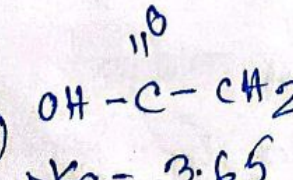
$$\Rightarrow 2 \log [H^+] = \log K_1 + \log K_2$$

$$\Rightarrow 2 pH_i = pK_1 + pK_2$$

$$\Rightarrow pH_i = \frac{pK_1 + pK_2}{2}$$

OR $$\boxed{pI = \frac{pK_1 + pK_2}{2}} \rightarrow \text{Isoelectic point}$$

Math (1)




$$pI = \frac{1.88 + 3.65}{2}$$

$$= 2.77$$

Scanned with CamScanner

(2) $H_3\overset{+}{N}-(CH_2)_5-\underset{\underset{+NH_3}{|}}{CH}-COOH$

$pK_b = 10.53$ $\quad$ $pK_b = 8.95$

$$pI = \frac{10.53 + 8.95}{2} = 9.74$$

(3) Glycine (2021)

$\overset{+}{N}H_3-CH_2-COO^-$

$pK_2 = 9.6$

$pK_1 = 2.4$

$$pI = \frac{2.4 + 9.6}{2} = 6$$

(2021)

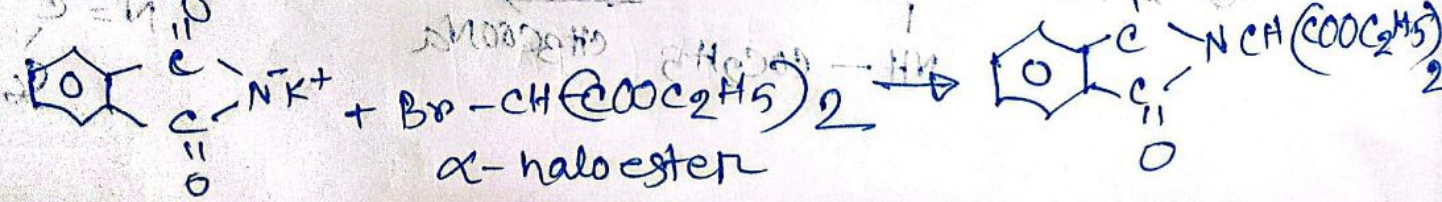
(x) synthesis of Amino Acid :-

(i) Direct Amination of α-halo acid

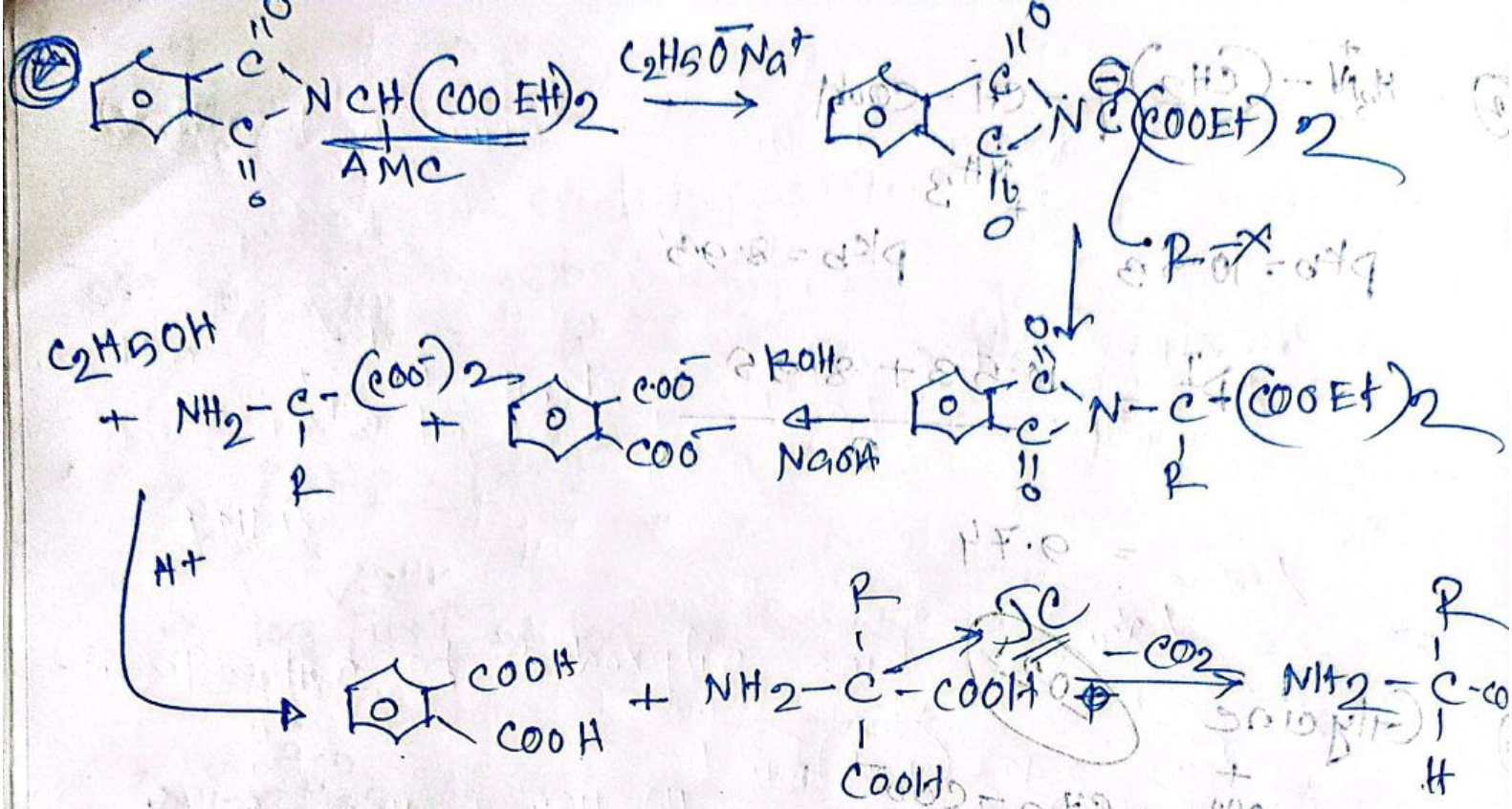
(ii) $R-CH_2-\underset{\text{F.Gr}}{COOH} \xrightarrow[\text{(ii) } H_2O]{\text{(i) } P+Br_2} R-\underset{\underset{Br}{|}}{CH}-COOH \xrightarrow[\text{Excess}]{NH_3} R-\underset{\underset{NH_2}{|}}{CH}-COOH$

50-60%

(ii) Gabriel phthalamide synthesis :-




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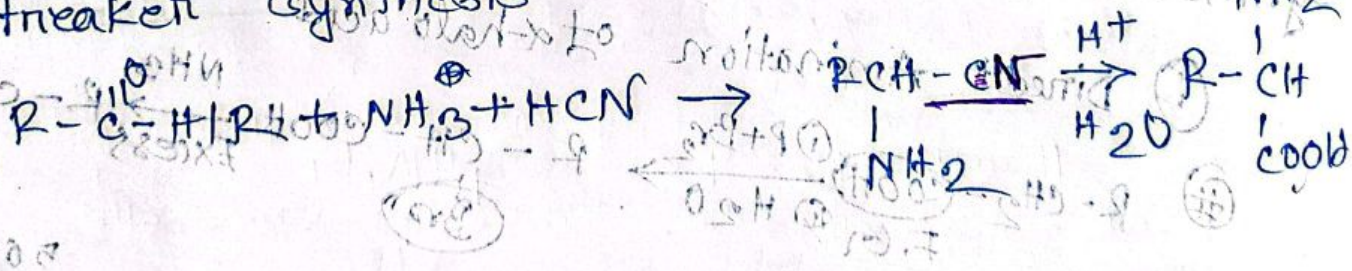



* Active methilin compound $(H^+)$ → High acidic
* Jeminal carbon (দুটি strong electron withdrowing group থাকে তাহলে stable হবে)

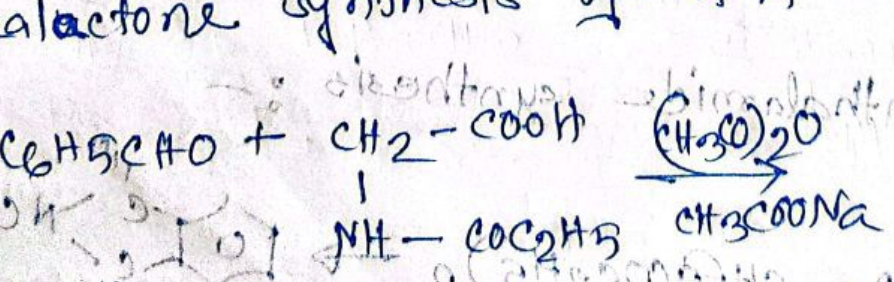
**Vicinal carbon**

**Streckar**

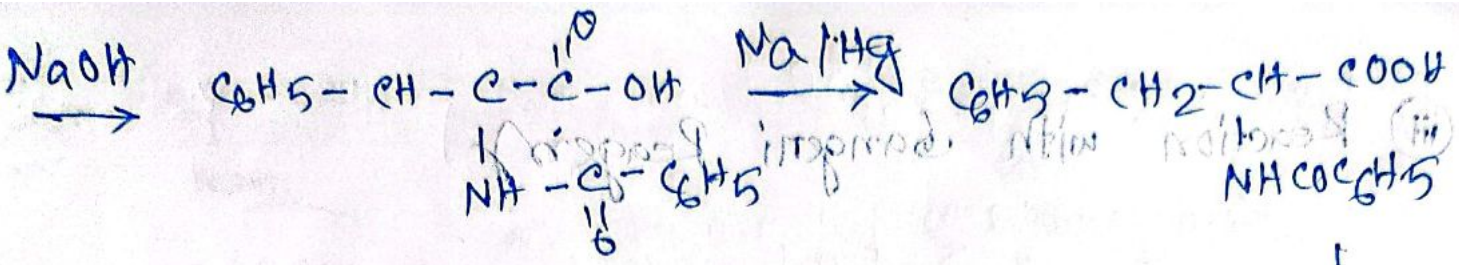
(iii) Streaker synthesis

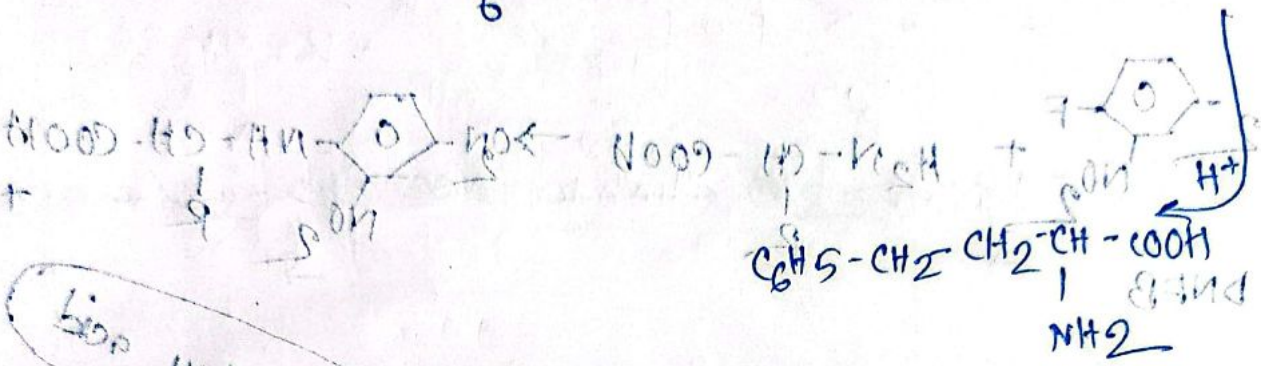



(iv) Azalactone synthesis of A.A.




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03-03-24- 4th class

(x) Reactions of α-amino acids:—

(i) $R-CH(NH_2)-COOH + (CH_3CO)_2O \rightarrow R-C(H)(NHCOCH_3)-COOH + CH_3COOH$

$R-CH(NH_2)-COOH + CH_3COCl \rightarrow R-C(H)(NHCOCH_3)-COOH + HCl$

Acetyl reaction

Drawback (inhale)

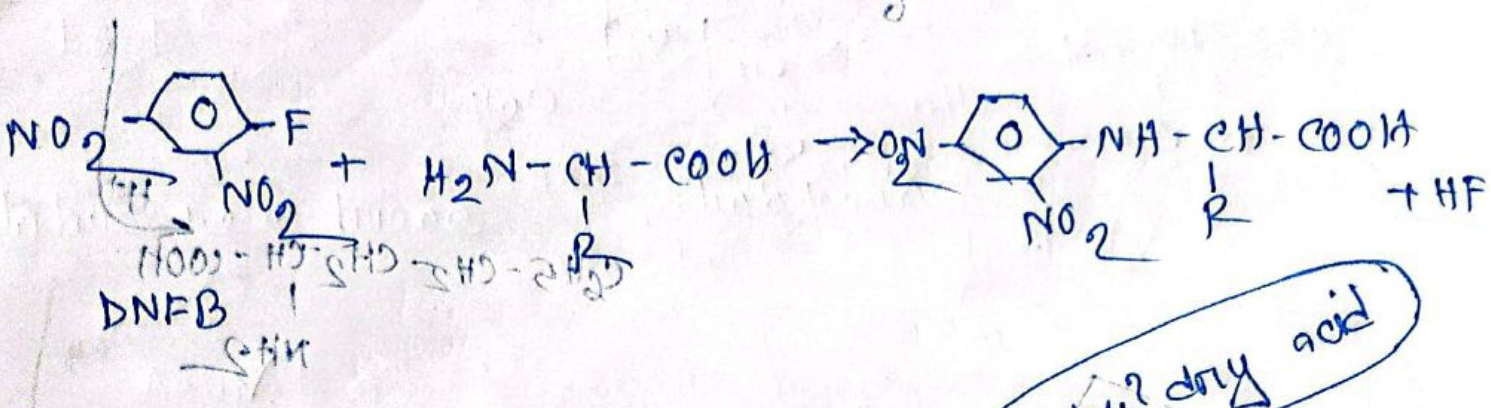
(ii) Alkylation 2022

$R-CH(NH_2)-COOH \xrightarrow[NaOH]{RX'} R-CH(NH(CH_2CH_3))-COOH$

$CH_3CH_2Cl$

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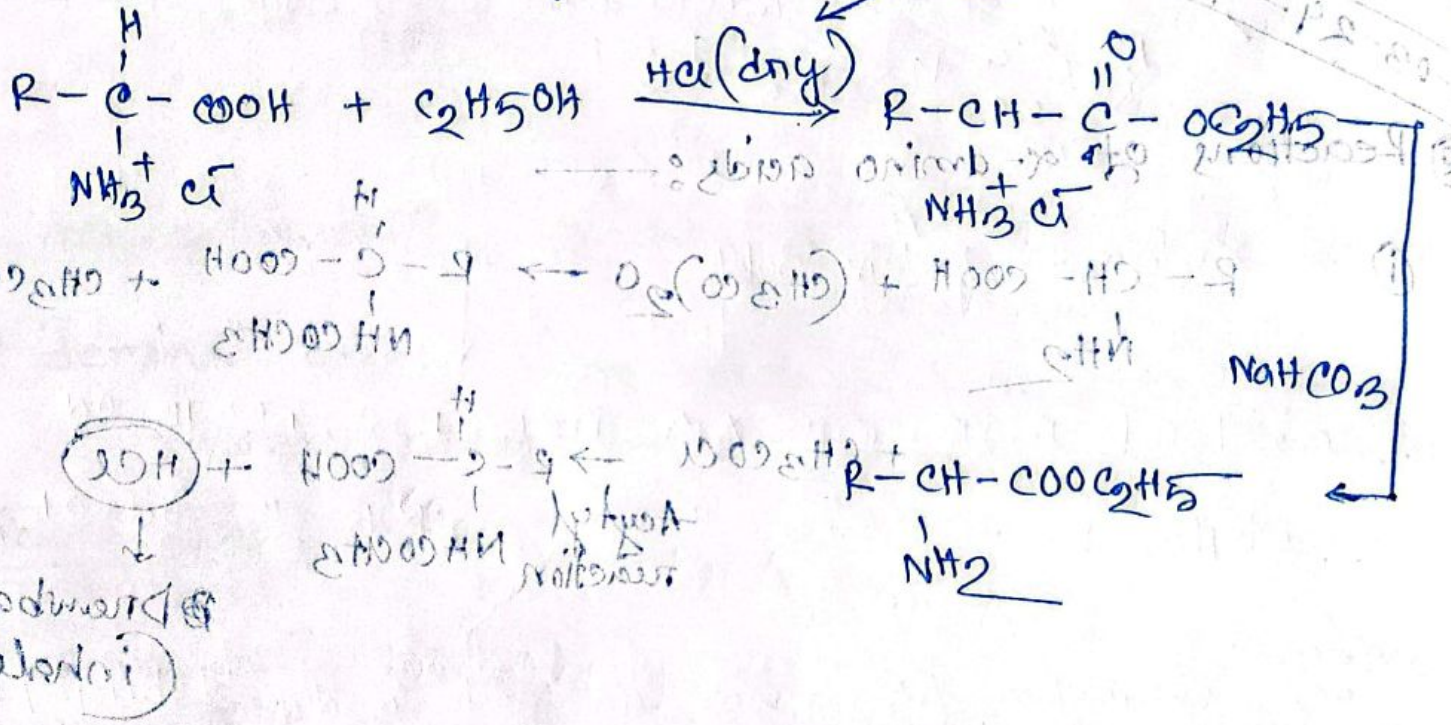
(iii) Reaction with Sangeri Reagent



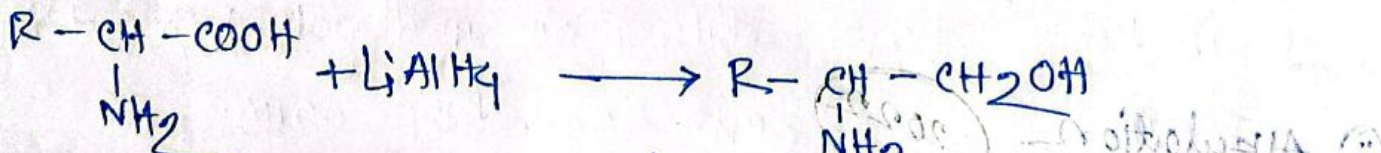
DNFB

(iv) Reaction of Carboxylic acid

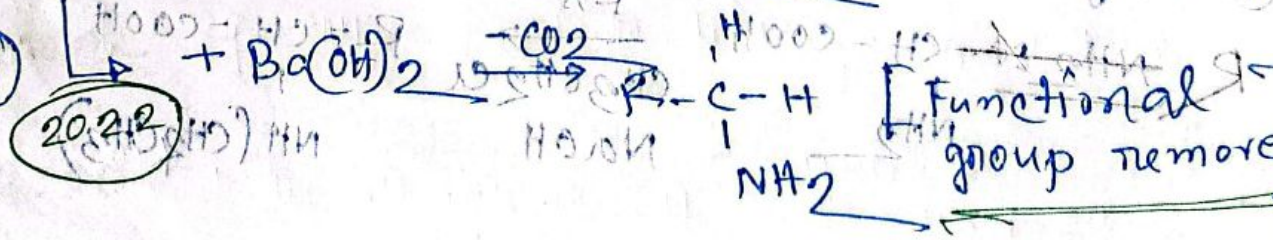
why? dry acid



(v)



(vi)



Functional group remove

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(7) 2021

$$R-\underset{NH_2}{\underset{|}{CH}}-COOH + HNO_2/HCl \xrightarrow{0-5^\circ C} R-\underset{OH}{\underset{|}{CH}}-COOH + N_2 + H_2O$$

α-hydroxy acid

(8)

$$HCHO + NH_2-\underset{R}{\underset{|}{CH}}-COOH \longrightarrow CH_2=N-\underset{R}{\underset{|}{CH}}-COOH$$

(9)

$$H_2C=O + NH_2-CH_2-COOH \longrightarrow CH_2=N-CH_2-COOH$$ 2021

(10)

$$R-\underset{NH_3^+}{\underset{|}{\overset{H}{\overset{|}{C}}}}-COO^- + NOCl \xrightarrow{pH=7} R-\underset{Cl}{\underset{|}{\overset{H}{\overset{|}{C}}}}-COO^-$$

(11) 2022

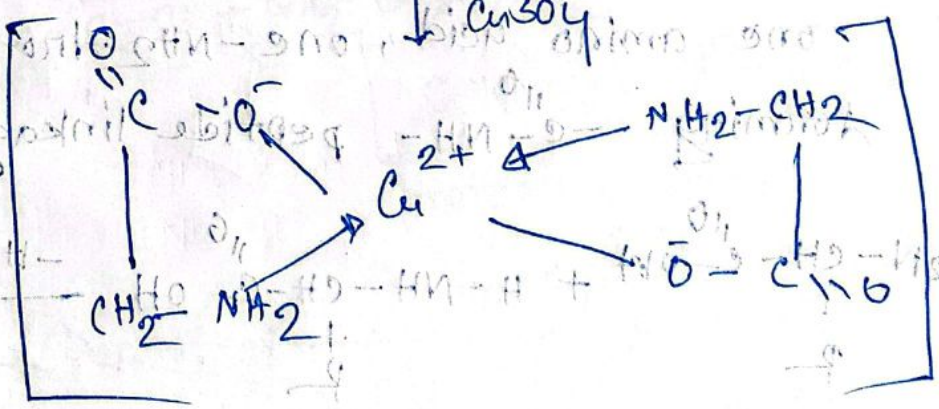
$$R-\underset{NH_2}{\underset{|}{CH}}-COOH + HI \xrightarrow[\text{Pyrolysis}]{200^\circ C} R-CH_2-COOH + NH_3$$

(*) Identification of α-amino acid 2022

(i)

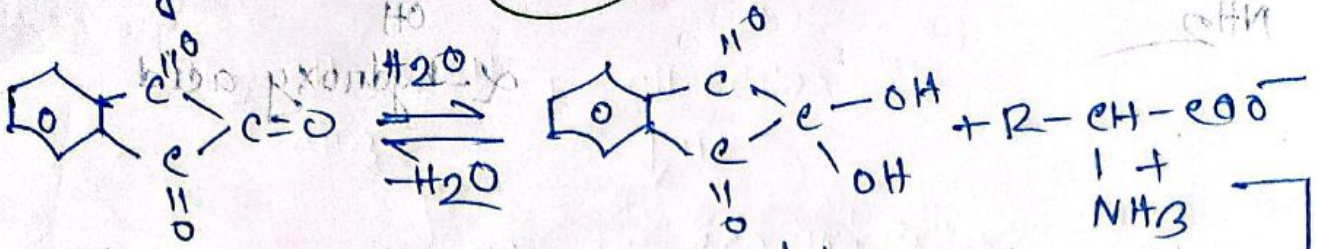
$$R/H-\underset{NH_3^+}{\underset{|}{\overset{H}{\overset{|}{C}}}}-COO^- \xrightarrow{H_2O} NH_2-CH_2-COO^- + H_3O^+$$

$\downarrow CuSO_4$

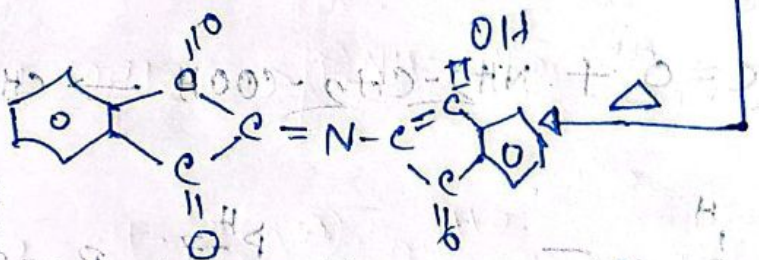


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(2) Ninhydrin test (2020)



Indane-1,2,3-trione Ninhydrine

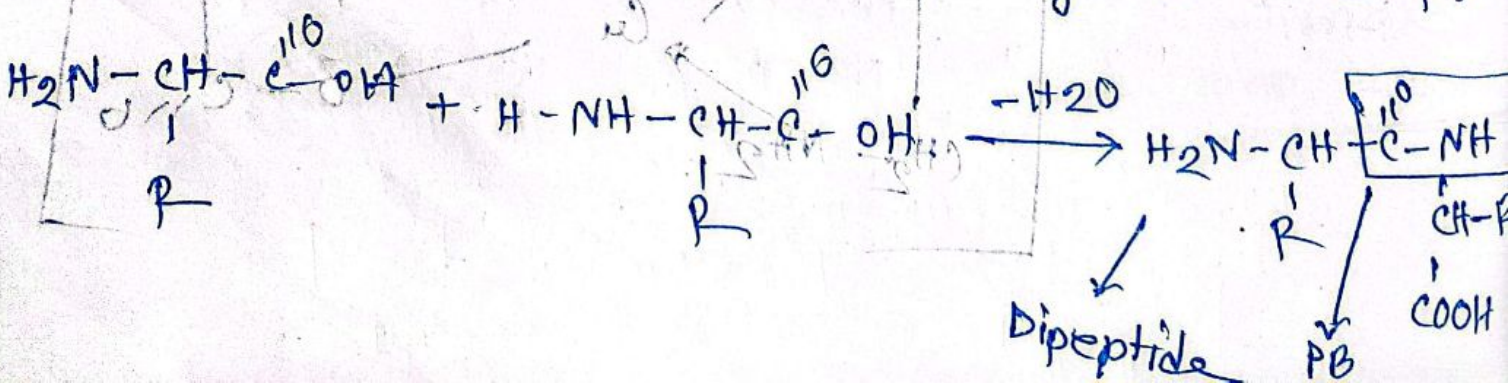


Blue-violet complex

50 বা এর চেয়ে বেশী AA যুক্ত হলে Protein.

10-50 " মধ্যে " " " Peptide.

(*) Peptides when two or more AA molecules are bonded together via peptide bond -COOH from one amino acid, one -NH2 from another forming $-C(=O)-NH-$ peptide linkage is called peptic




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$H_2N-\underset{H}{\underset{|}{CH}}-COOH$ + $NH_2-\underset{CH_3}{\underset{|}{CH}}-COOH$ + $NH_2-\underset{CH_2OH}{\underset{|}{CH}}-COOH$

Glycine Alanine Tyrosine

04-03-24 – 5th class

(2021)

(*) synthesis of peptide:

$NH_2-CH_2-COOH + NH_2-\underset{CH_3}{\underset{|}{CH}}-COOH \rightarrow NH_2-CH_2-\overset{O}{\overset{||}{C}}-NH-\underset{CH_3}{\underset{|}{CH}}-COOH$

Glycine or Alanine

$NH_2-\underset{CH_3}{\underset{|}{CH}}-\overset{O}{\overset{||}{C}}-NH-\underset{CH_2PH}{\underset{|}{CH}}-\overset{O}{\overset{||}{C}}-NH-CH_2-COOH$

Alanine Tyrosine Glycine

Alanyltyrosyl Glycine

Ala-Tyro-Gly

$NH_2-CH_2-CONH-CH_2-COOH$ or

$NH_2-\underset{CH_3}{\underset{|}{CH}}-CONH-CH_2-COOH$ or

$NH_2-CH_2-CONH-\underset{CH_3}{\underset{|}{CH}}-COOH$ or

$NH_2-\underset{CH_3}{\underset{|}{CH}}-CONH-\underset{CH_3}{\underset{|}{CH}}-COOH$

(*) ① Protection of $NH_2$

$C_6H_5-CH_2-O-\overset{O}{\overset{||}{C}}-Cl + NH_2-CH_2-COOH \xrightarrow{Base} C_6H_5-CH_2-O-\overset{O}{\overset{||}{C}}-NH-CH_2-\overset{O}{\overset{||}{C}}-OH$

Protecting group

Amide neutral, reactivity low

Deprotect $\xrightarrow{H_2/Pt/Pd}$

$C_6H_5-CH_2-H + HO-\overset{O}{\overset{||}{C}}-NH-CH_2-COOH \xrightarrow{-CO_2} NH_2-CH_2-COOH$

Scanned with CamScanner

② Protection of $-COOH$ group:-

$NH_2-CH(CH_3)-COOH + OH-CH_2C_6H_5 \xrightarrow{H^+}$

$NH_2-CH(CH_3)-COOCH_2C_6H_5$ (Protected)

Activation of $-COOH$ group

$NH_2-CH_2-COOH + SOCl_2 \rightarrow NH-CH_2-CO-Cl$ (P)

$NH-CH_2-CO-Cl + NH_2-CH(CH_3)-COOCH_2C_6H_5$ (P$_1$) $\xrightarrow[\text{(ii) Deprotect}]{\text{(i) HCl}}$ $NH_2-CH_2-CO-NH-CH(CH_3)-COOH$

P$_1$ = Protected

N terminal, C-terminal

③ Solid phase method to synthesis glycylalanine

2020 (C)

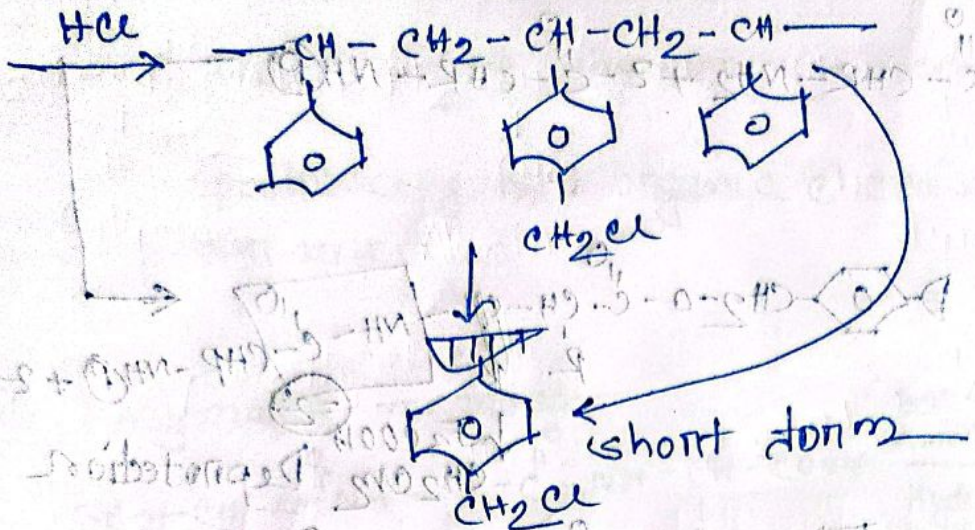
(i) Solid phase

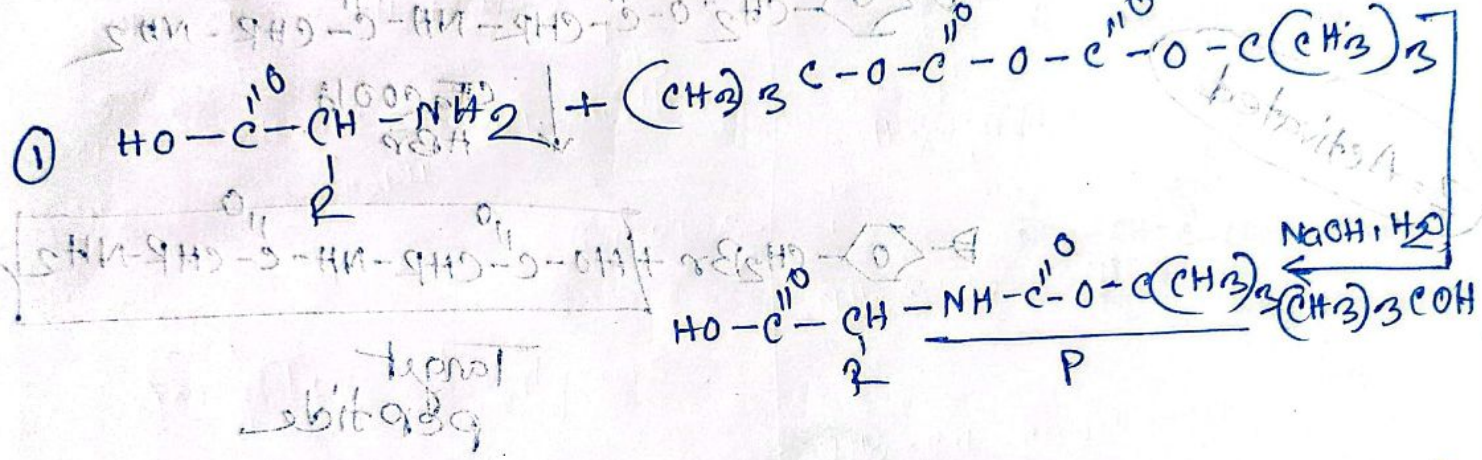
$-CH(C_6H_5)-CH_2-CH(C_6H_5)-CH_2-CH(C_6H_5)-$ $\xrightarrow[HCl]{HCHO}$ Polystyrene

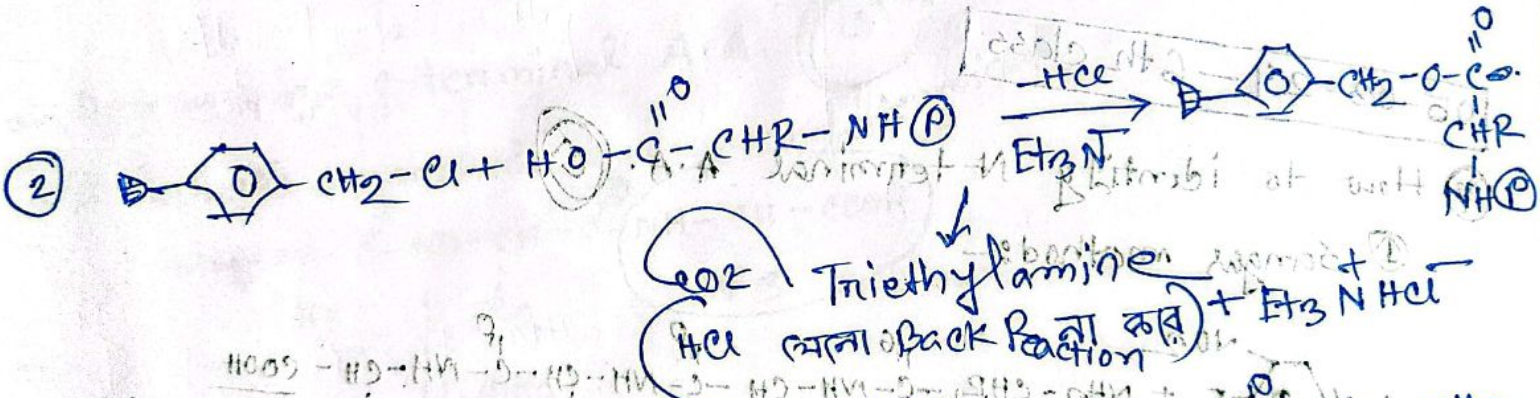
$CH_2=CH(C_6H_5)$ Styrene

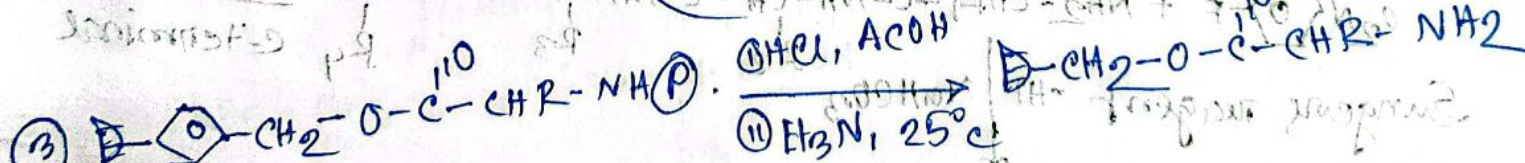
$-CH(C_6H_5)-CH_2-CH(C_6H_5)-CH_2-CH(C_6H_5)-$

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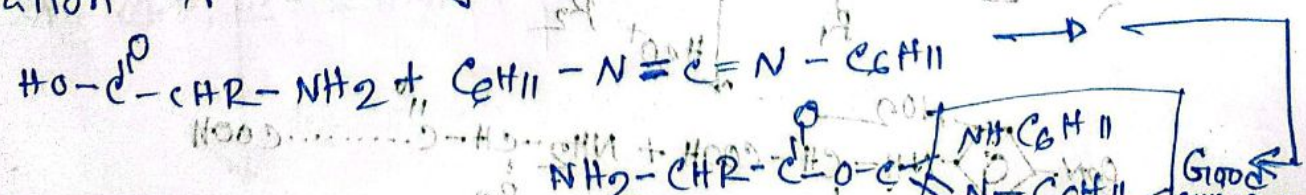


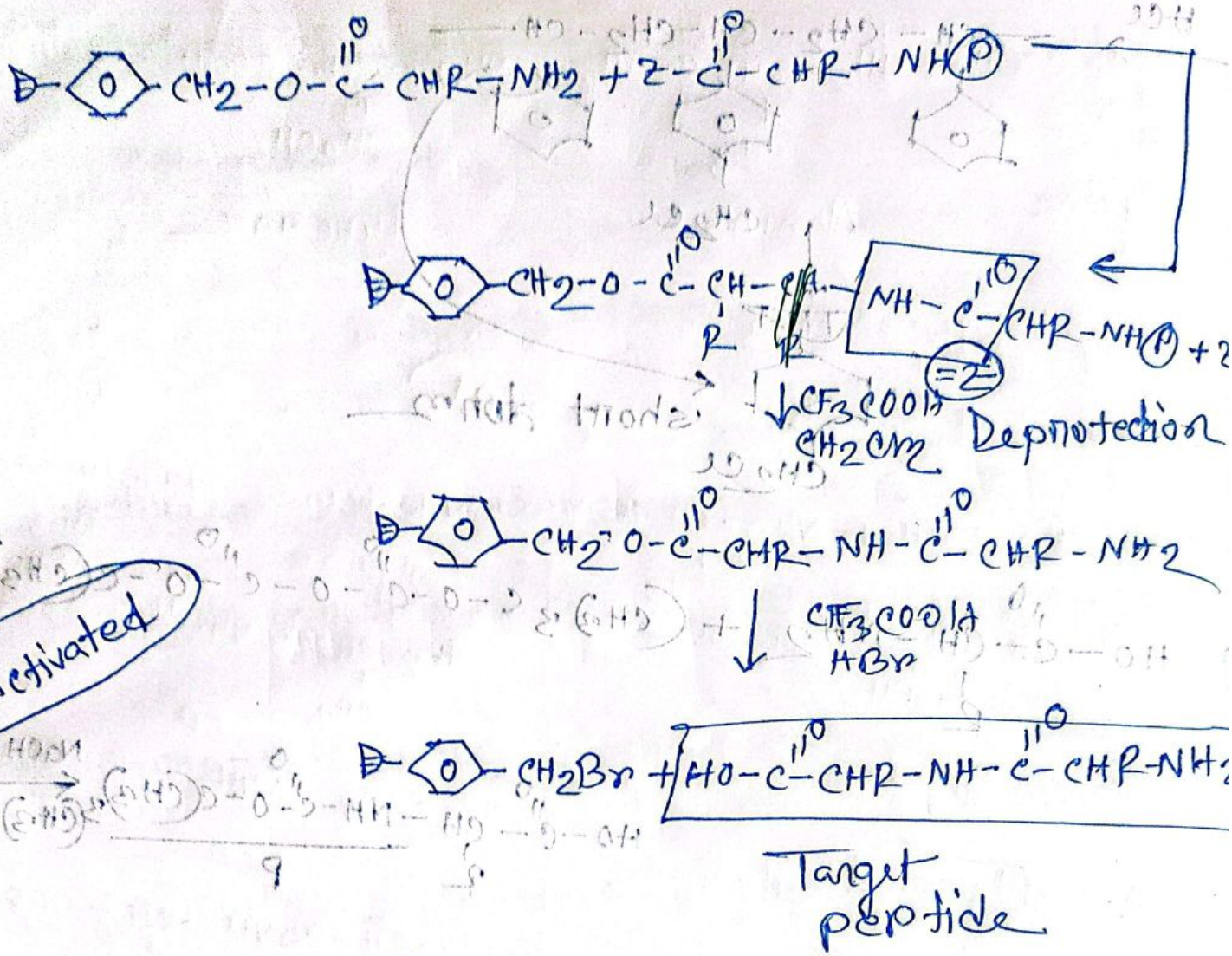




(4) Activation of -COOH group




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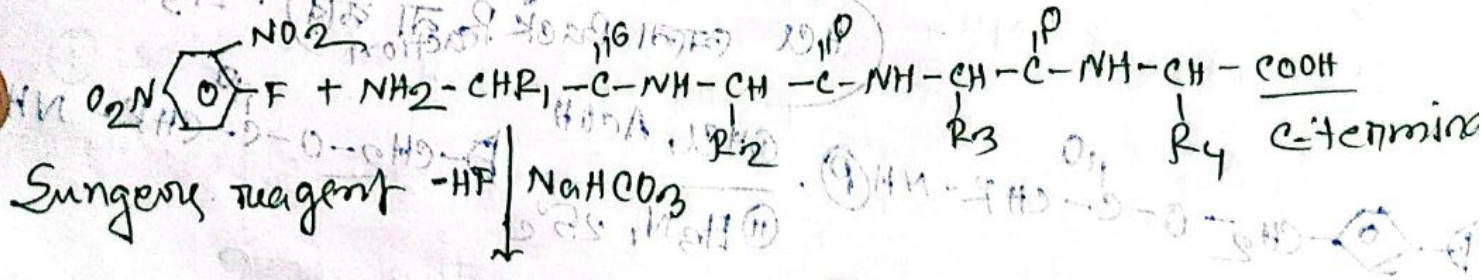
⑤ 

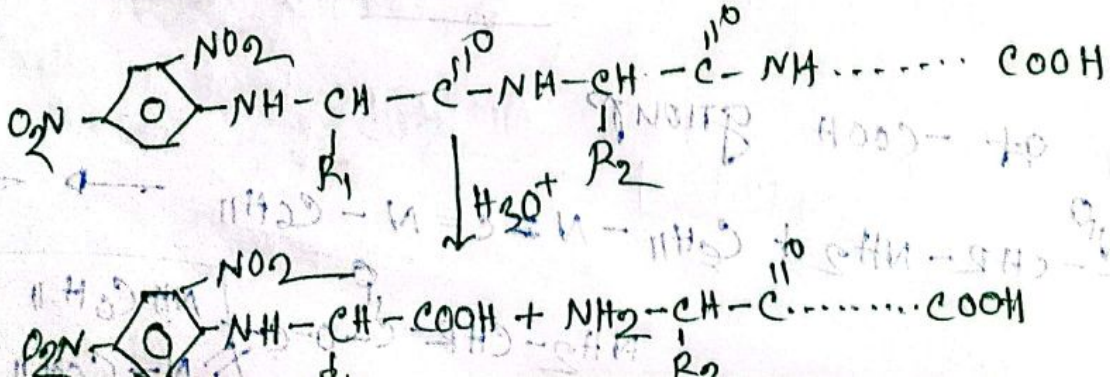

Z = Activated

05-03-24 — 6th class

⑥ How to identify N-terminal A.A. (CT)

① Sangus method:—

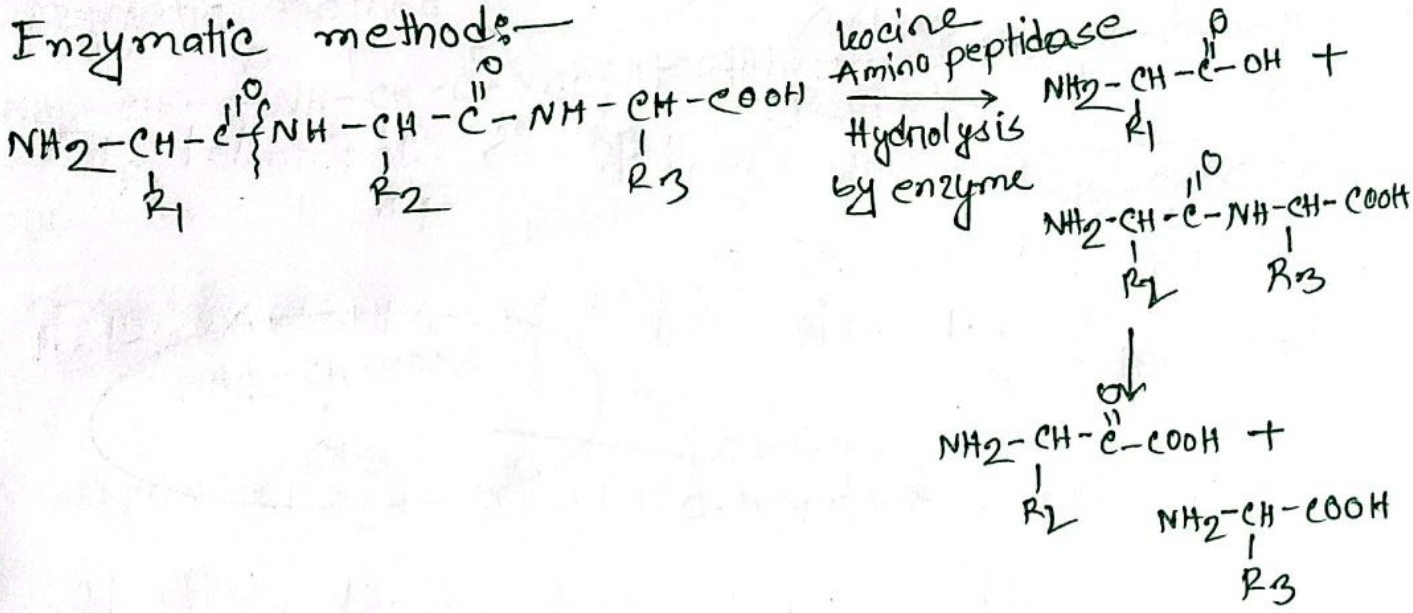



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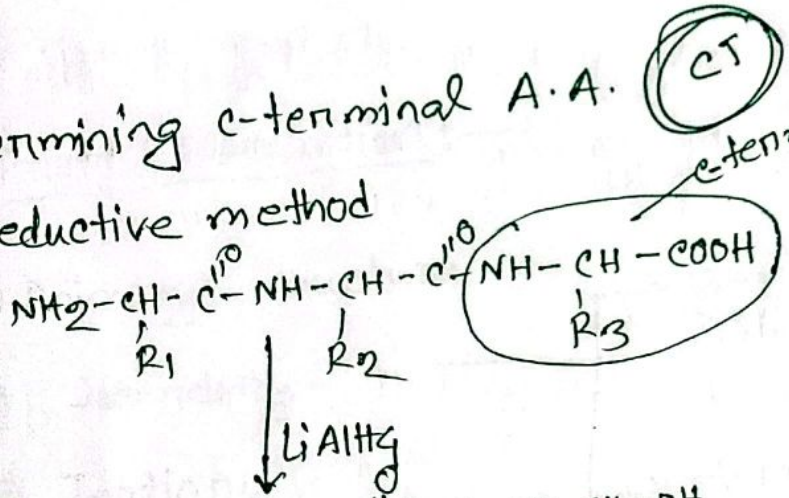
যে A.A এর sangers reagent যুক্ত থাকবে তাহলে N-terminal spectroscopy দ্বারা organic chemistry structure বের করা যায়।

② Enzymatic method:—




# Determining c-terminal A.A. (CT)

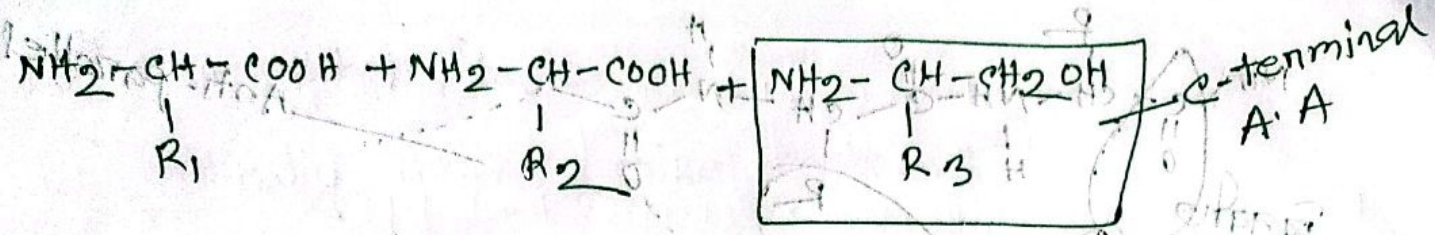
① Reductive method



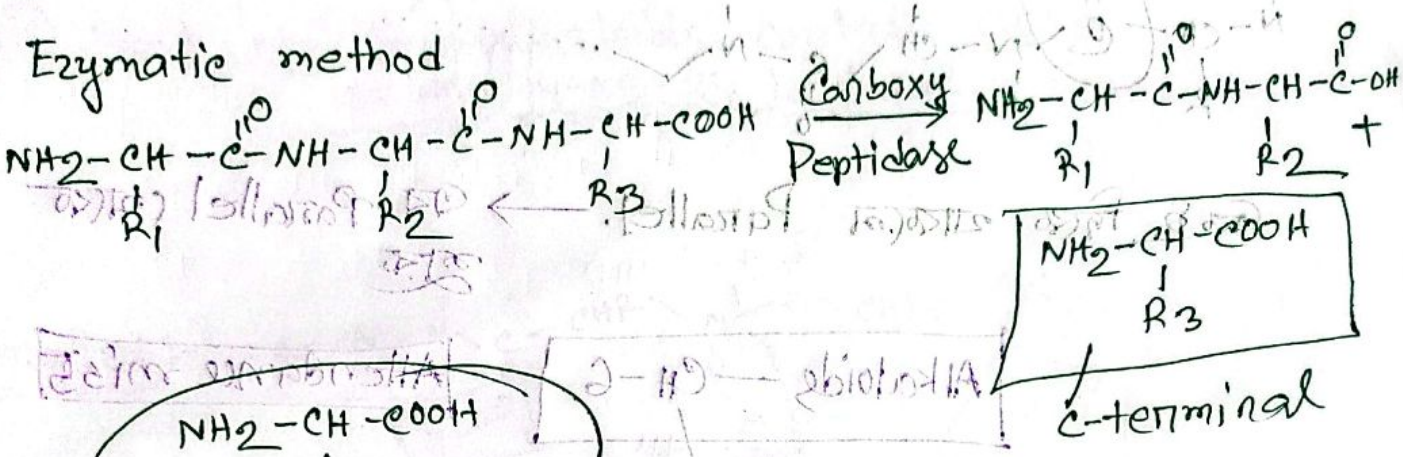

যে A.A এর অগ্রে Alcohol থাকবে Hydrolysis করার পর তাহা c-terminal.

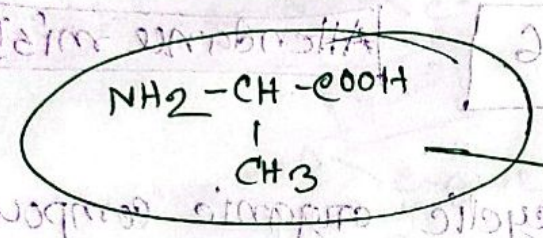
Hydrolysis

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② Ezymatic method



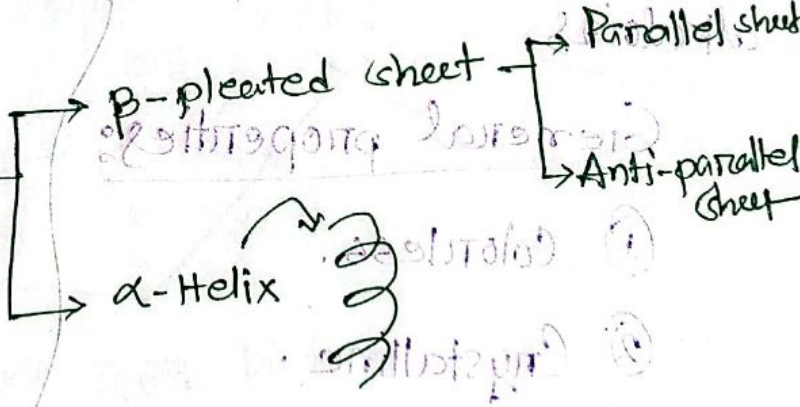

④ Protein Definition:

① Primary structure

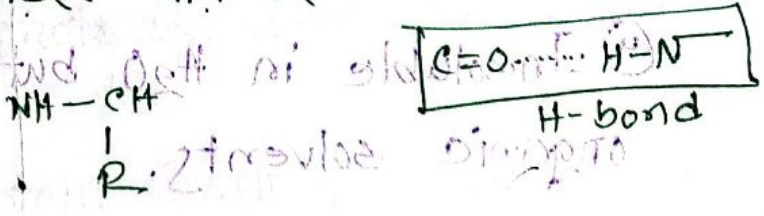
② Secondary "

③ Tertiary "

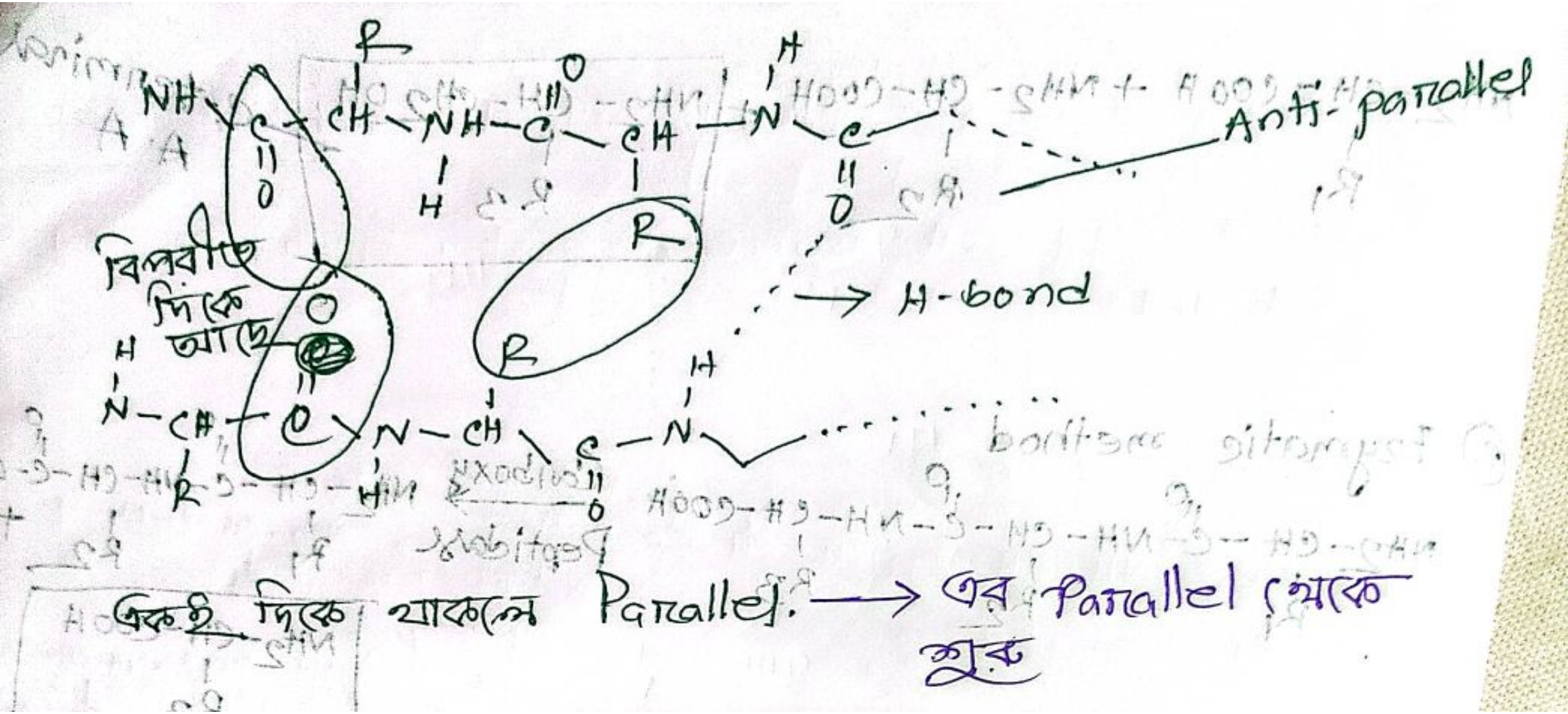
④ Quaternary "




R বড় হলে দূরত্ব বাড়বে যার ফলে H-bond গঠন করতে পারবে না।

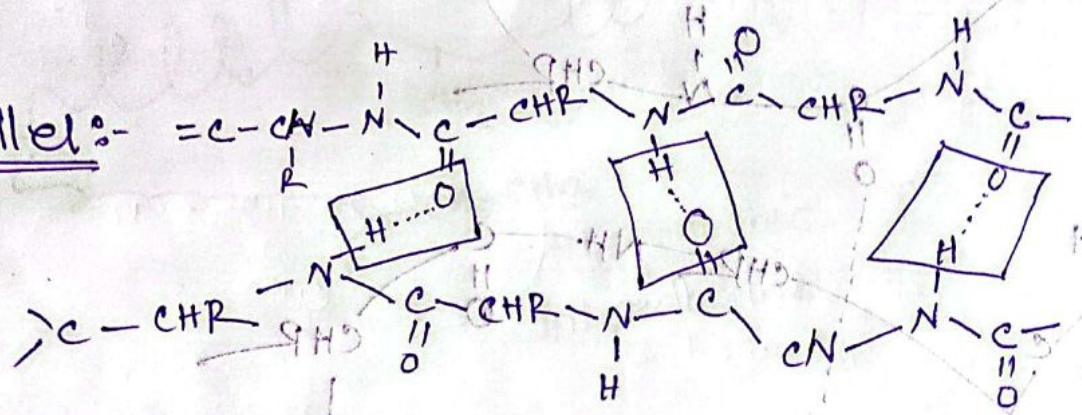



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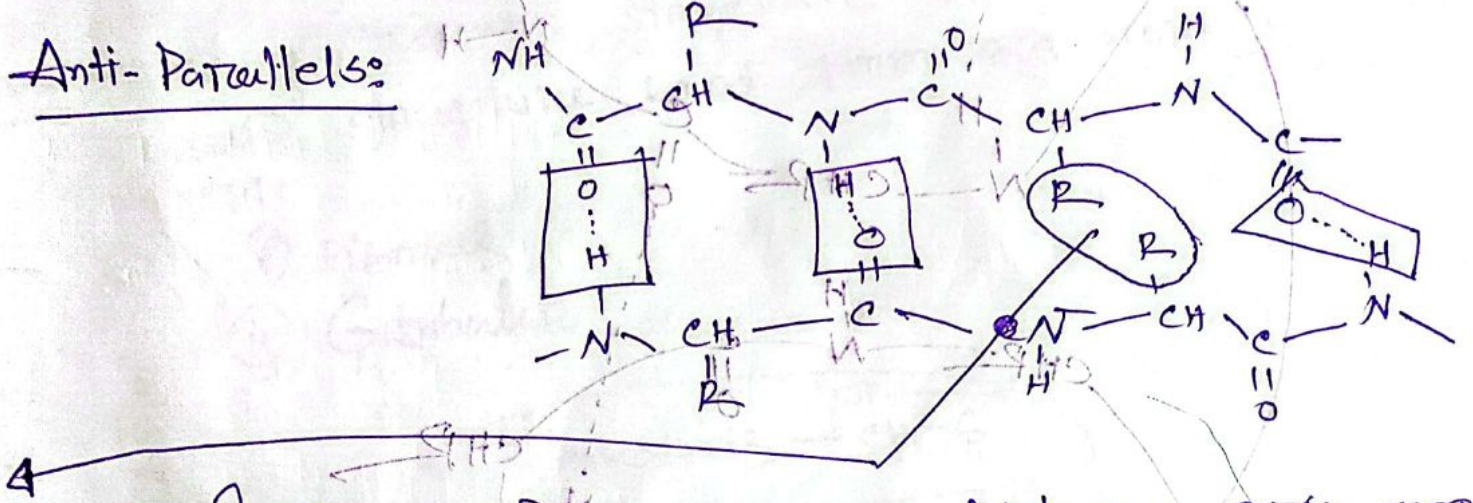



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**Parallel:-**



**Anti-Parallels:**



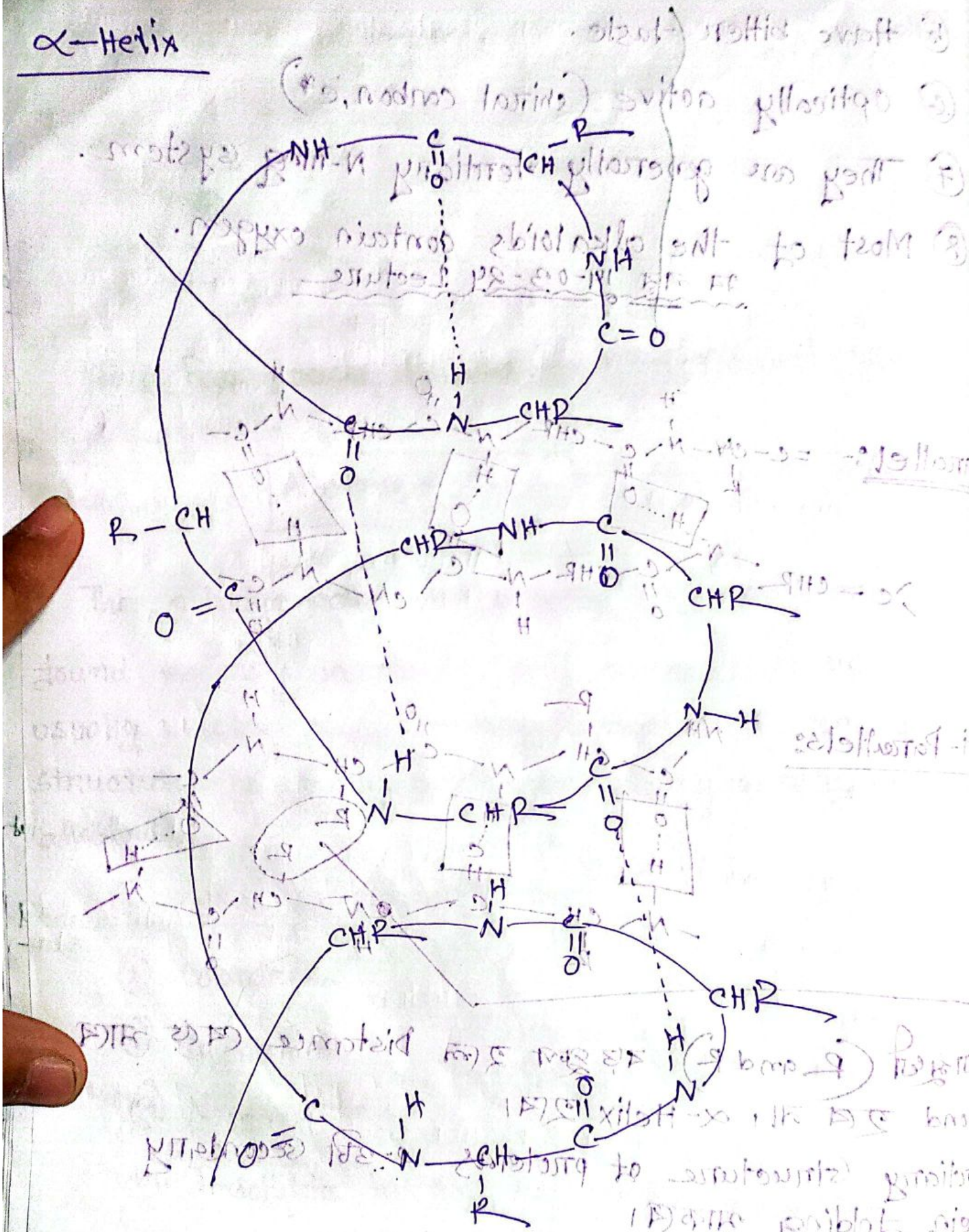
মুখোমুখী (R and R) বড় গ্রুপ হলে Distance বেড়ে যাবে।
H-bond হবে না। α-Helix হবে।

⇒ Tertiary Structure of proteins ২-টা secondary protein folding থাকবে।

**Quaternary:** ২-৩টা Protein একসাথে interaction হবে

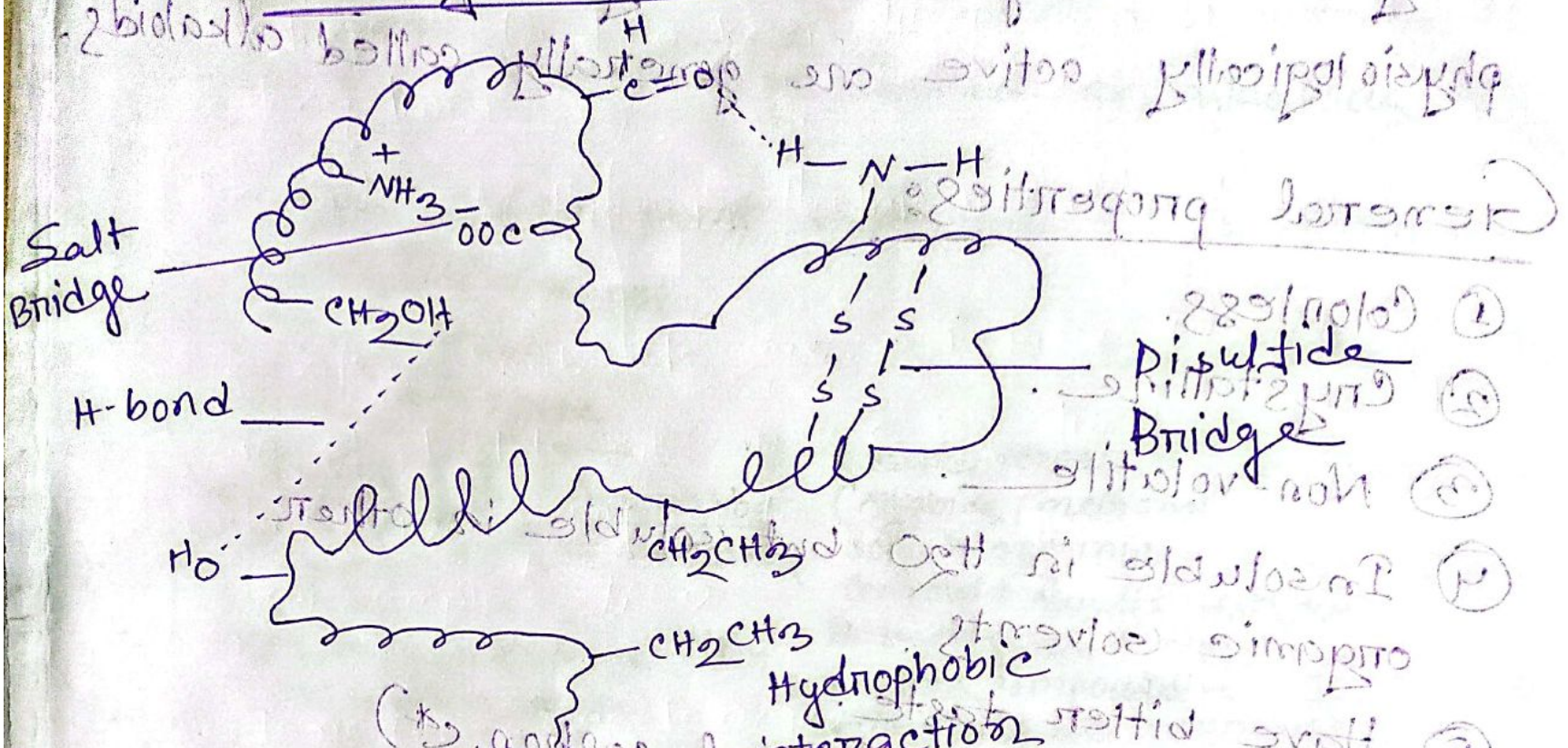
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# α-Helix




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## Tertiary Structure




## Quaternary Structure

একটি structure bond form করে যখন Quaternary হবে।

(i) Fibrous

(ii) Globular